HT66FB574/572 Colour Effect USB Mouse Application Note

D/N：AN0483E

Introduction

Demands from the video gaming industry for different types of gaming mouse continue to expand. Adding a large number of RGB LEDs to the mice can produce different colours and brightness changes creating a range of visual special effects. This enhances the colour and stimulating effects of gaming mice. For example, having multiple RGB LEDs to form an outer ring on a gaming mouse can produce a colour changing waterflow effect. These are known as colour effect USB mice.

Firmly directed at the multiple colour RGB LED product application area, Holtek has three USB RGB Flash MCUs, the HT66FB572, HT66FB574 and HT66FB576, which have the ability to control up to 40/64/128 RGB LEDs. Their integrated USB interface makes these devices very suitable for colourful keyboard and mouse products. This application note will introduce how to use the HT66FB574/572 devices to develop colour effect mice.

Specification Description

In this application note, the colour effect USB mouse specification is as follows.

(1) USB Full Speed.

(2) 6 buttons: Left/Middle/Right/Forward/Back/Wheel buttons.

(3) Wheel×1.

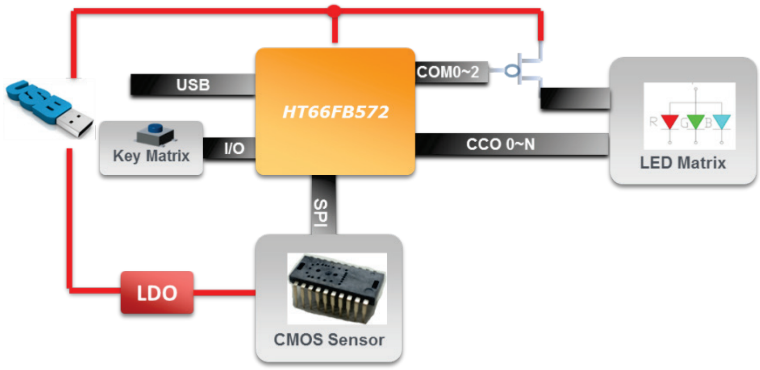
(4) Provides 6 lighting effects controlled by software.

(5) Resolution 1000/1500/2000/2500/3000 DPI set by software with the maximum DIP determined by the CMOS Sensor.

(6) Report rate: 125/250/500/1000Hz.

(7) Supported operating systems: Windows XP, Windows Vista, Windows 7, Windows 8 and Windows 10.

Hardware Block Diagram



HT66FB572 Colour Effect USB Mouse Hardware Block Diagram

Operating Principle

USB 2.0 Full Speed Interface

The HT66FB57x series provide a USB interface function, making them suitable for general computer peripheral and consumer product applications. To communicate with an external USB host, the internal USB module has three external pins known as UDP and UDN along with a 3.3V regulator output V33O. The USB interface has 8 endpoints, EP0~EP7. All endpoints except endpoint 0 can be configured with an 8, 16, 32 and 64 bytes FIFO size using the UFC0~UFC2 registers in the application programs. Endpoint 0 has an 8 byte FIFO size. Endpoint 0 supports control transfer, while endpoints 1~7 support interrupt or bulk transfer.

USB function control is implemented using a series of registers. A series of status registers provide the user with the USB data transfer information as well as error conditions. The USB interface contains its own independent interrupt which can be used to indicate when the USB FIFOs are accessed by the host device or a change of the USB operating conditions including the USB suspend mode, resume event or USB reset occurs.

Refer to the USB interface section in the corresponding datasheet for more information.

RGB LED PWM Driver

The HT66FB57x series include an integrated constant current source, which together with their 15, 24 and 48 PWM output pins, have the ability to control up to 40, 64 and 128 RGB LEDs in a matrix scan configuration. The flexible RGB LED colour change possibilities can implement breathing and lighting effects using fully internal hardware circuits. This makes these highly integrated MCUs suitable for use in colour RGB LED products such as RGB Gaming Keyboards, RGB Gaming Mouse, RGB LED Speakers, etc. These devices reduce the need for an external PWM IC resulting in lower costs and reduced product development time.

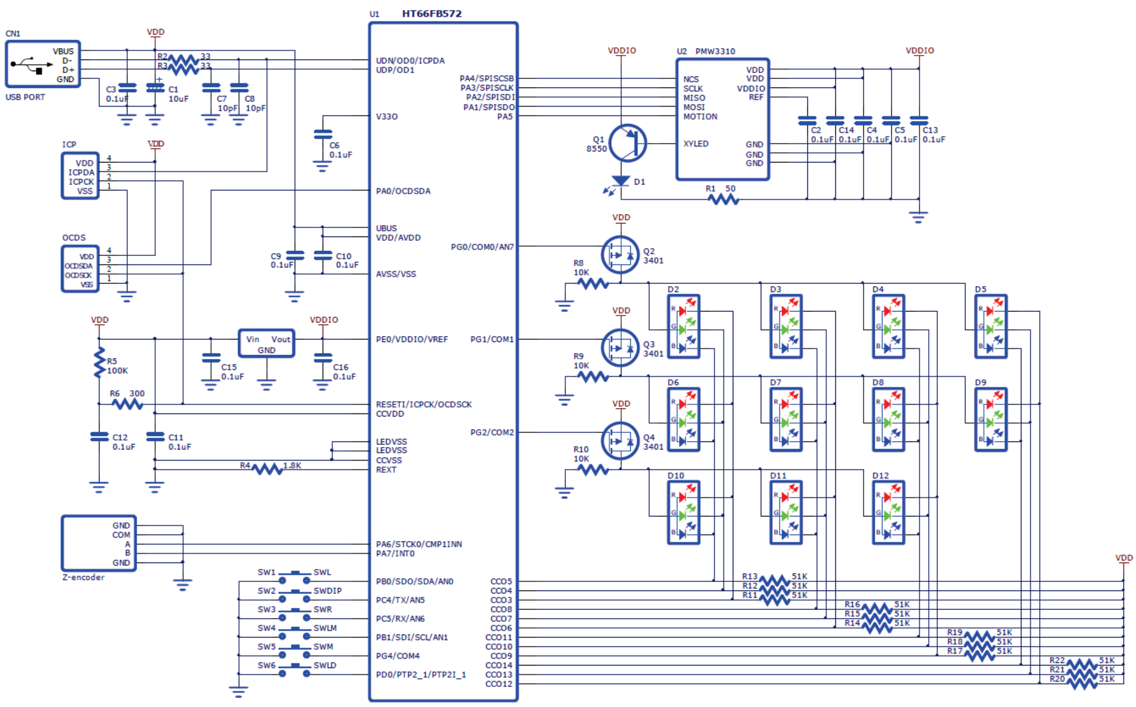
The HT66FB57x each includes two LED PWM Data Memory sectors. The hardware can automatically calculate the PWM duty cycle according to the data written into a dedicated Data Memory area to implement LED colour changes and breathing effects. For more details, refer to the relevant section in the HT66FB57x Data Sheet.

Refer to the RGB LED section for details on the LED PWM operations.

CMOS Sensor

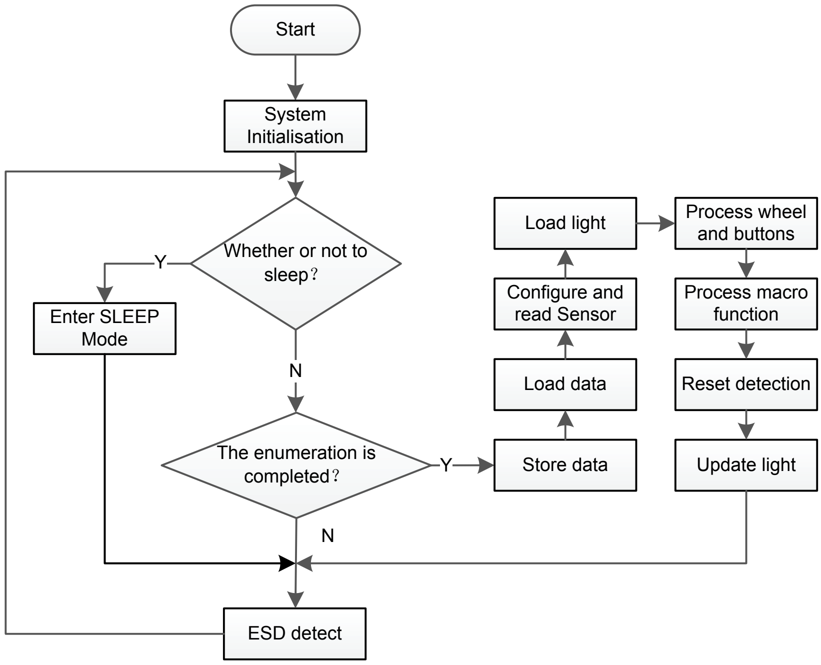
The CMOS Sensor operating voltage is 3.3V in this application note. The HT66FB57x series provide an SPI interface for communication with a CMOS Sensor. It also provides a VDDIO function to allow the SPI interface to have an equivalent voltage level interface with the CMOS Sensor, which is more convenient to use.

Hardware Circuit

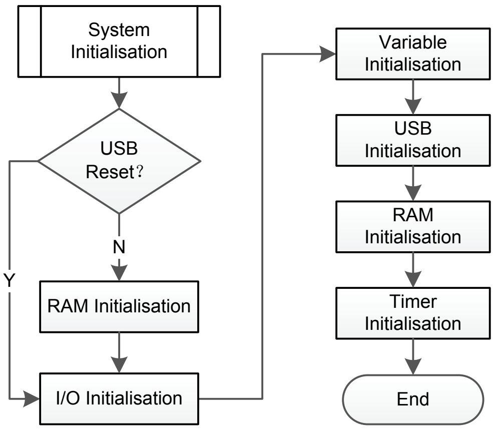


HT66FB572 Colour Effect Mouse Circuit Schematic Diagram

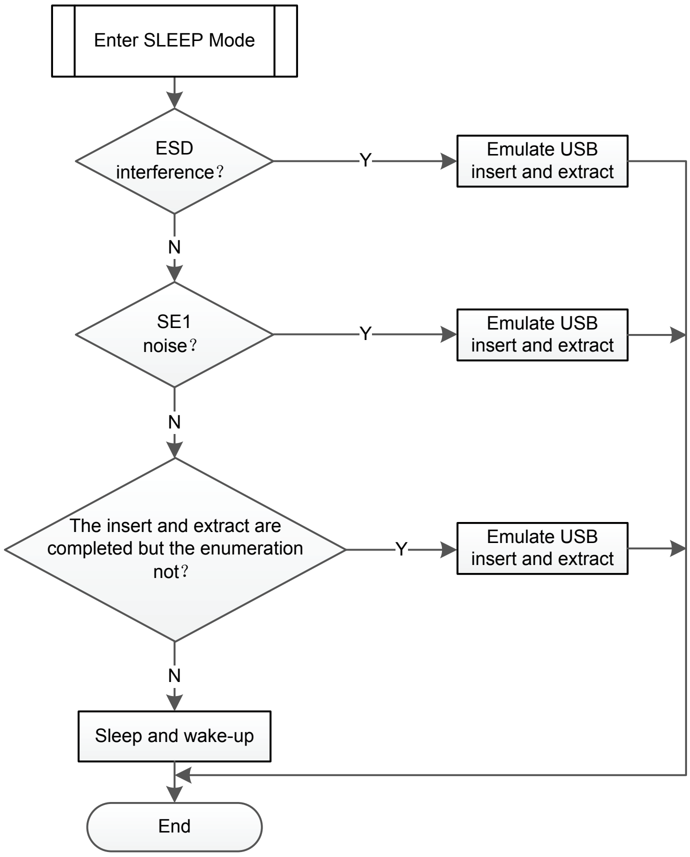
Program Description



Main Program Flowchart

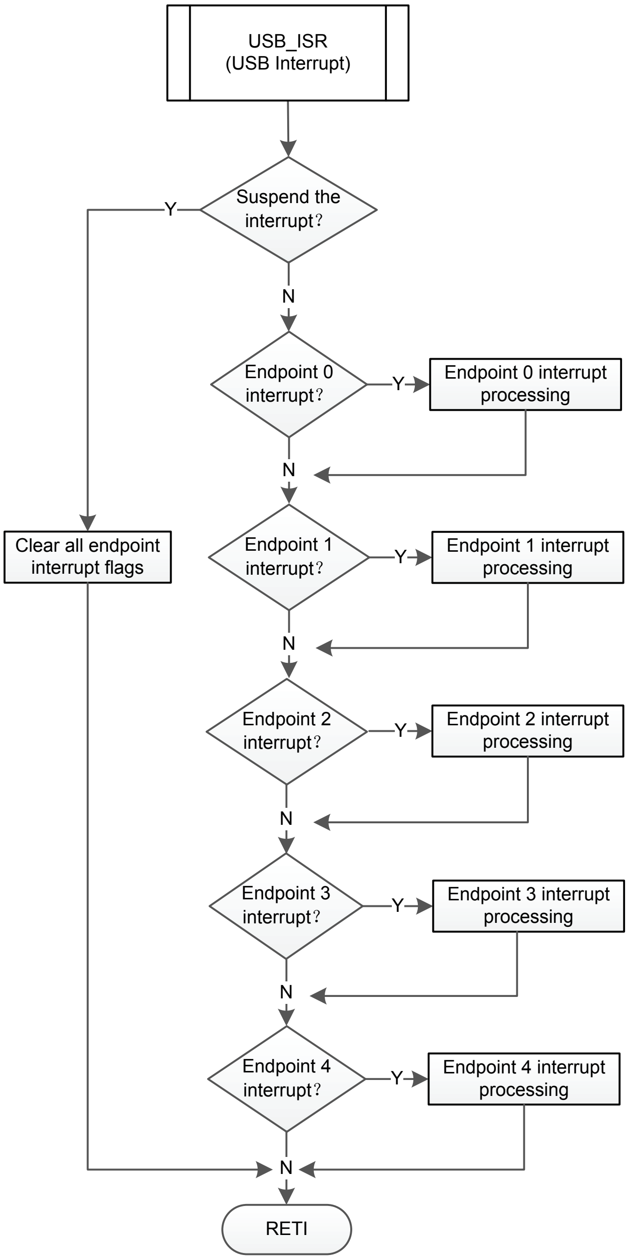


System Initialisation Subroutine Flowchart



Sleep Subroutine Flowchart

Note: The above actions can improve the mouse anti-interference ability. When an interference signal is detected, the USB insert and extract operations will be emulated by software. When the interference has ceased the enumeration will be restarted.



USB Interrupt Subroutine Flowchart

After the USB function starts, a USB interrupt will be generated when the USB is inserted. In this application note, Endpoint 0 is used for enumeration or for some report commands. Endpoint 1 is used to transmit data related to the mouse. Endpoint 2 is used to transmit data related to the keyboard. Endpoint 3 and endpoint 4 are used for bulk transfer data set by the host computer AP.

AP Operation Description

Mouse Outline

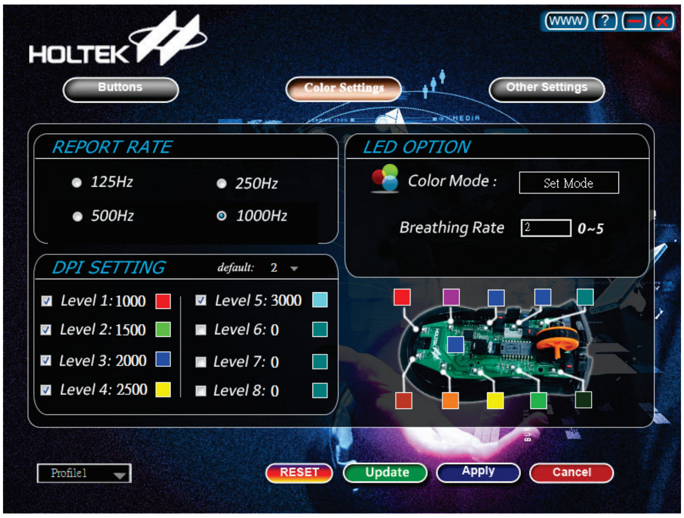
User-defined button – available for the HT66FB574

Users can use a powerful AP driver to define 8 buttons, which are double-click, menu, multimedia, text office etc. In addition, the DPI switch button combined with the forward button is used to switch the lighting effect mode. The DPI switch button combined with the back button is used to adjust the lighting effect and breathing rate.

AP Driver Button Page

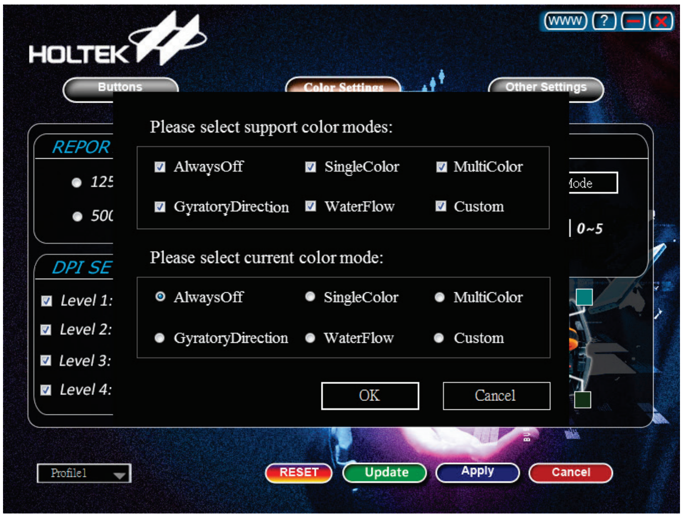
On this page, users can define the button functions. After the update is completed, click on the "Apply" button. The “RESET button is used to reset the default button function. The “Update” button is used to update the mouse MCU software.

User-defined Report Rate and Lighting Effect – available for the HT66FB574/572



AP Driver Colour Setting Page

On this page, users can setup the report rate, DPI level and LED colour, colour mode and breathing rate. After clicking “Set Mode”, the following window will pop up to allow selection the supported colour modes and current colour modes.



AP Driver Set Mode Window

There are a total of six colour mode types: AlwaysOff, SingleColor, MultiColor, GyratoryDirection, WaterFlow, Custom.

1. SingleColor：selectable breathing rate (0~5), user-defined colour.

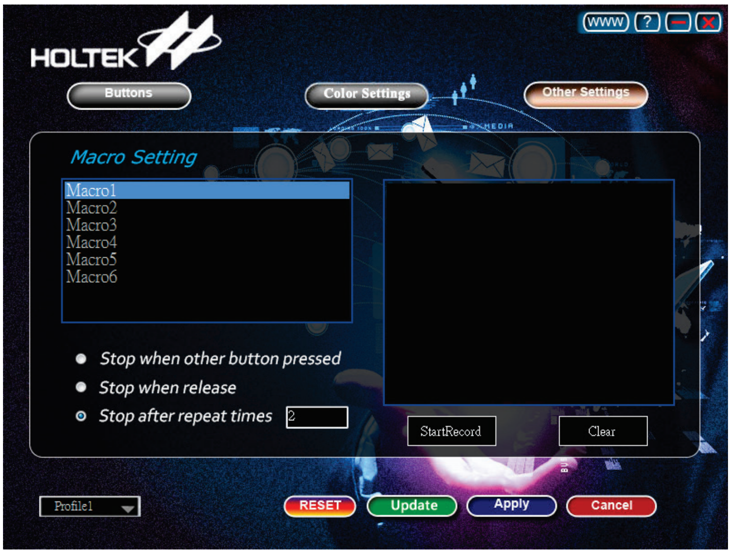
2. MultiColor：selectable breathing rate (0~5), random colour.

3. GyratoryDirection：selectable breathing rate (0~5).

4. WaterFlow：selectable breathing rate (0~5), user-defined colour.

5. Custom: users can define the colour and whether each LED breathes normally.

Macro Definition – available for HT66FB574



AP Driver Other Setting Page

On this page, users can setup the Macro function. There are three types of macros, which are stop when another button is pressed, stop when released and stop after repeat times

There are six profiles, profile0~5. Profile1~5 each has 6 macros, so there are 30 macros in total. The macro action can be a combination of a series of common mouse buttons and keyboard buttons, which can record up to 31 button actions. The maximum action interval is 131 seconds.

Program

Refer to the colour effect mouse application solution on Holtek's website for the related AP and F/W program attachments.

Conclusion

The HT66FB572 and HT66FB574 devices include an integrated LED PWM constant current control circuit. They have the ability to control up to 40 and 64 RGB LEDs in a matrix scan configuration together with two internal RGB LED PWM Data Memory sectors. It is not difficult to implement different kinds of lighting effects and with no requirement for an external LED driver IC and no requirement for a large number of transistors, the result is lower costs and reduced product development time.

With a USB 2.0 Full Speed interface, these are excellent solutions for applications which require a large amount of RGB LEDs such as USB Gaming Mice.

Version and Modify Information

Date	Author	Issue and Revision
2018.2.14	Wang Guanzhong(王冠中)	First Version

Reference Files

1. Refer to the HT66FB576 Data Sheet.
2. HT66FB576_574_572 RGB LED USB MCU PWM function application note.

For more information, refer to the Holtek's official website http://www.holtek.com

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